

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

06th November 2015

Subject ST6 – Finance and Investment B

Time allowed: Three Hours (10.15* – 13.30 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You have then three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet unless instructed to do so by the supervisor.*
4. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question paper and no page is missing. If so, kindly get a new set of Question paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** An insurance company is exposed to falls in the swap spread of term 10 years (difference between swap rate and government bonds rate) over the next year. An investment bank approaches the insurance company to hedge this risk by using total return swaps and forward contracts on bonds.
- i)** Describe total return swaps and forward contracts on bonds. (3)
 - ii)** Consider that today you are at $t=0$. Explain, using cash flows at $t=1$, how a combination of total return swaps and forwards on bonds can be used to capture movement of swap spreads? Specify details of the hedge positions. (8)
 - iii)** What are the key risks that the insurance company would be exposed to if it decides to enter in this structured transaction and how these risks could be managed? (4)
- [15]**

- Q. 2)** An insurance company purchases equity-linked bonds of term 5 years for Rs 1 crore which promises to pay Rs 1 crore at maturity plus a coupon based on the increase in the value of the NSE Nifty index with participation rate of 100%.

The government is proposing new regulations to calculate capital charges on these notes. The charge is based on the following

- 1) 5% of the fixed interest rate exposure
- 2) A capital charge for the options to be calculated as equal to the maximum of the fall in the option prices based on four scenarios generated by the changes in two assumptions (all combinations of rise and fall for both risks)

Volatility	NSE Nifty Index
15% increase	25% increase
15% decrease	25% decrease

- i)** The level of the NSE Nifty index at the date of transaction is 8000 and the current volatility is 25%, the risk free interest rate is 5%, the risk premium for the company is 2%. Estimate the capital charge. State any of the assumptions used. (8)
 - ii)** Comment on the suitability of the bond for the insurance company if the cost of capital for the insurance company is 3.5% over the risk free rate and suggest alternate assets to gain similar exposures (3)
- [11]**

- Q. 3)**
- i)** Describe the Merton model and the principle of using the firm's asset value for pricing the firm's bonds. Also state the assumptions underlying the model. (5)
 - ii)** Use the Merton model to calculate the value of a security to be issued by the company and its spread over the risk free rate. In this model, a non-dividend paying company issues one single bond: a zero-coupon bond. (5)

Value of the firm	100
Face value of bond	60
Maturity (years)	10
Risk-free interest rate	5.00%
Volatility of firm's assets	30.00%

[10]

- Q. 4)** It has been observed that the probability distribution of stock prices is not exactly lognormal.

Discuss issues with using the Black-Scholes formula for pricing call and put options on assets whose values have different tail distribution compared to lognormal (though with same mean and variance).

Illustrate your answer using distribution diagrams.

[6]

- Q. 5)** A student actuary at Limca Plc suggested a new interest rate model which is driven by the following rule:

- 1) The interest rate goes up and down with factor $\text{Sqrt}(\text{rate}(t-1) \times \text{fact1})/10$ each year, where $\text{rate}(t)$ is specified as percent value, that is, 3.1% means 3.1.
- 2) The probability of interest rates going up and down is 50%.

The student introduced an approximation to build a binomial tree which was that every possible bridging node is derived from the previous step higher node, calibrating to historical rates using $\text{fact1} = 3$.

Calculate, using the above model, the value of a three-year 3.0% cap and floor with annual resets. The current interest rate is 3.0% and the notional principal is Rs 1 crore.

[8]

- Q. 6)** Consider shares of two companies, both of which are positively correlated with the performance of the NSE Nifty index and depend on no other stochastic variables. Suppose that the first one has an expected return of 20% per annum and a volatility of 25% per annum, and the second one has a volatility of 35% per annum. Assume that the instantaneous risk free rate of interest is 6% per annum. What is the market price of risk for NSE Nifty index? What is the expected return from the second security?

[6]

- Q. 7)** You are an investment actuary offering consulting services specializing in derivatives. Suppose it is October 1, 2015 today and a client comes to you for advice. He is holding 1000 shares of Actuarial Corporation of Utopia (ACU) and he has observed that the futures are trading at a substantial discount to the market. The client wants to sell the stock and buy the futures.

Before you offer your advice or proceed further, you do some research on ACU prices and observe the following (as of October 1, 2015):

- ACU spot price quotes (Bid/Ask): Rs. 650 / Rs. 652
- ACU 1 month futures (i.e. expiring on October 31, 2015) quotes (Bid/Ask): Rs. 590 / Rs. 594
- Interest rate quotes for all terms (Bid/Ask): 5% / 6%
- Dividend per share: Rs. 45
- Ex – Dividend Date: 15th October 2015
- Brokerage on Spot Transaction: 1% of the spot value
- Brokerage on Futures Transaction: 0.5% of the future price
- Margin required at any point in time: 5% of the underlying value as at the close of the day (assume Rs. 651)

Further, for the purpose of this question assume that futures will NOT be settled daily and margin will be released at the maturity along with the settlement.

Based on the above information,

- i) What will be your advice with respect to the soundness of the client's strategy of selling the stock and buying futures? (9)
 - ii) Also, what are the risks that the client will be exposed to while implementing the strategy? (5)
- [14]

Q. 8) Consider a perpetual bond with a continuously compounded yield of “ y ”. Further, the bond pays interest at Rs. 2 per annum. Assume that interest is paid continuously on the bond, and that “ y ” follows the process:

$$dy = \mu (\alpha - y) dt + \sigma y dZ$$

where μ , α and σ are positive constants and dZ is a standard Brownian motion.

Given this information, answer the following two questions:

- i) What is the process followed by the bond price? (6)
 - ii) What is the expected return to the bond holder expressed as a percentage of the bond price? (4)
- [10]

Q. 9) The table shows the bid – ask quotes (Bid/Ask) by Alpha Bank (AB) for Credit Default Swap (CDS) spreads for companies A, B and C.

	1 year	3 years	5 years
A	15 / 25	21 / 32	27 / 36
B	43 / 60	72 / 101	112 / 152
C	71 / 84	93 / 113	141 / 170

Beta Limited has excessive credit exposure on company C and wants to reduce the same through the CDS market. Since the farthest maturity of its exposure to C is 3 years, Beta Limited buys \$200 million 3 year protection on C from AB. In order to reduce its cost of buying the protection, based on its views on company A and B, Beta Limited decides to sell \$300 million 5 year protection on company A and \$100 million 1 year protection of company B to AB. What is the net annual premium payment made by Beta Limited to AB in the first year? [5]

Q. 10) ABC Limited has issued a loan of \$400 million at 6.5% per annum. ABC Limited enters into a Total Return Swap (TRS) under which it shall pay the interest on the loan plus the change in the mark – to – market value of the loan. In exchange, ABC Limited will receive LIBOR + 50 bps. Settlement payments are made **semi – annually**. What is the cash flow for ABC Limited on the first settlement date if the mark – to market value of the loan has fallen by 2% and the LIBOR is 4%. [5]

- Q. 11)** Consider the following homogeneous reference portfolio in a synthetic Credit Default Swap (CDS):

The number of reference entities = 100

CDS spread, $s = 150$ basis points

Recovery rate, $f = 50\%$

Also, assume that the defaults are all independent of each other. On a single name, the annual default probability is constant over five years and obeys the relation

$$s = (1 - f)PD$$

where PD is the probability of default.

What is the expected number of defaulting entities over the next five years, and which of the following tranches would be entirely wiped out (lose 100% of the principal invested) by the expected number of defaulting entities?

[6]

- Q. 12)** Suppose that yield Y on a zero-coupon bond follows the process:

$$dY = mdt + \sigma dz$$

where m and σ are functions of Y and t , and dz is a wiener process.

How would the volatility of the zero-coupon bond price behave as the bond gets closer and closer to maturity?

[4]
